

SOFTWARE REQUIREMENTS SPECIFICATION

SHRAVU – Invisible by Choice. Connected by Permission

App Name: SHRAVU

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INTRODUCTION

1.1 Purpose

A Privacy-first P2P messaging application built natively for India's 2026 DoT SIM Binding Regularity Framework. SHRAVU Implements a “Ghost by Default” Identity model where users are invisible unless they actively choose to connect. The document serves as the autoreactive specification for all development, testing, and compliance activities.

Core Objectives:

Objective	Description
Ghost-by-Default Identity	Users are unsearchable. Connections happen only via single-use, rotating "Handshake" codes to ensure total public invisibility.
HW-Bound Accountability (DoT 2026 Compliance)	One Human, One ID. Uses mandatory SIM-binding (IMSI/IMEI) for 2026 DoT compliance and to eliminate bot accounts.
Absolute Data Revocation (The Poison Kill)	A "Clean Break" tool that remotely and irreversibly wipes shared chat history from both devices using DOD-standard zero-filling.
Decentralized Resilience (Cloud-Less Recovery)	No cloud storage. Chat history is reconstructed directly from trusted peer contacts, maintaining a pure zero-knowledge architecture.
Temporary Professional Service Exchange	Temporary professional channels with auto-collapse timers and usage counters to prevent work access from becoming permanent.

1.2 Intended Audience

Category	Role
Engineering	Frontend Engineers (React Native / Flutter)
Engineering	Backend Engineers (Node.js/Rust)
Engineering	Cryptographic Engineers
Quality & Security	QA & Security Auditors
Legal	Legal & Compliance Officers
Product	Product Management

1.3 Product Scope

In Scope	Out of Scope: (for now)
Ghost Handshake connections (Daily QR + Service ID QR)	International SIM support
P2P + relay messaging	Public channels / broadcasts
Group management with GSK key ratchet	
Poison Kill	
Peer-assisted recovery	
Subscription management	
File sharing beyond images and video	

1.4 Intended Use

- This document is to be used as:
- The primary engineering handoff document for development
- The basis for test plan creation
- The compliance reference for regulatory audits.

1.5 Definitions and Acronyms

Term	Definition
SIM Binding	Hardware-level linking of a physical SIM (IMSI + IMEI) to an app account instance.
Ghost Handshake	Proprietary QR-based P2P connection protocol where QRs rotate daily and burn on scan.
Tunnel	Encrypted bilateral P2P communication channel between two SHRAVU IDs
Connection ID	Ephemeral identifier used in Ghost Handshake; rotates every 24 hours for personal, configurable for service
GSK	Group Session key – shared symmetric cryptographic key for group message decryption
KDF	Key Derivation Function – one-way mathematical function (e.g. HKDF-SHA256) used in GSK
Poison Kill	SHRAVU’s irreversible connection termination mechanism; zero-fills local storage on both devices and blacklists the Connection ID on server
Zero-Fill	Overwriting storage sectors with binary zeros to prevent forensic recovery
Relay Blob	Encrypted message payload stored temporarily on server when receiver is offline
Service Taker	User who generates a Service ID QR to invite professional service providers
DRN	Designated Relay Node – the group member (admin-first) who routes group messages
Epoch	A GSK version period; each GSK rotation creates a new epoch
IMSI	International Mobile Subscriber Identity – hardware identifier embedded in SIM card
IMEI	International Mobile Equipment Identity – unique hardware identifier of the device
Dot	Department of Telecommunications, Government of India
DPDP	Digital Personal Data Protection Act 2026

1.6 References

- DoT Telecommunication Cybersecurity Amendment Rules, 2025 (effective March 1, 2026)
- Digital Personal Data Protection Act (DPDP), 2026
- Information Technology Act, 2000 (India), Sections 69 & 79
- Signal Protocol Specification (Open Whisper Systems) - reference for E2E encryption patterns
- IETF RFC 5869 — HKDF Key Derivation Function Specification
- SHRAVU Architecture Whitepaper v1.1 (Internal Document)
- SHRAVU Privacy Policy v2.0 (Legal Document)

OVERALL DESCRIPTION

2.1 Product Perspective

SHRAVU is a greenfield product with no predecessor. It operates as a standalone mobile application (Android/iOS), backed by three server components:

Relay Server — Stores encrypted blobs for offline delivery

Compliance Server — Stores metadata logs for legal compliance

Signalling Server — Facilitates P2P connection establishment

At no point does message content reside on the server in plain text. The system is designed from the ground up to comply with India’s 2026 DoT SIM Binding rules, making SHRAVU natively compliant, whereas competitors are retrofitting compliance.

2.2 Product Features

One Number, One Identity: Phone number → unique @name#number ID. Phone number is invisible to other users.

Ghost by Default: No global search, no contact sync; invisible unless explicitly connected.

Ghost Handshakes: Daily rotating QR (personal) + Service ID QR (professional, time-limited).

P2P + Ephemeral Relay: Direct device-to-device when both are online; encrypted relay when offline (text: 30 days, media: 48 hours).

Group Key Ratchet: Group Session Key rotates automatically on every membership change. Perfect forward secrecy. Max 2 keys on device.

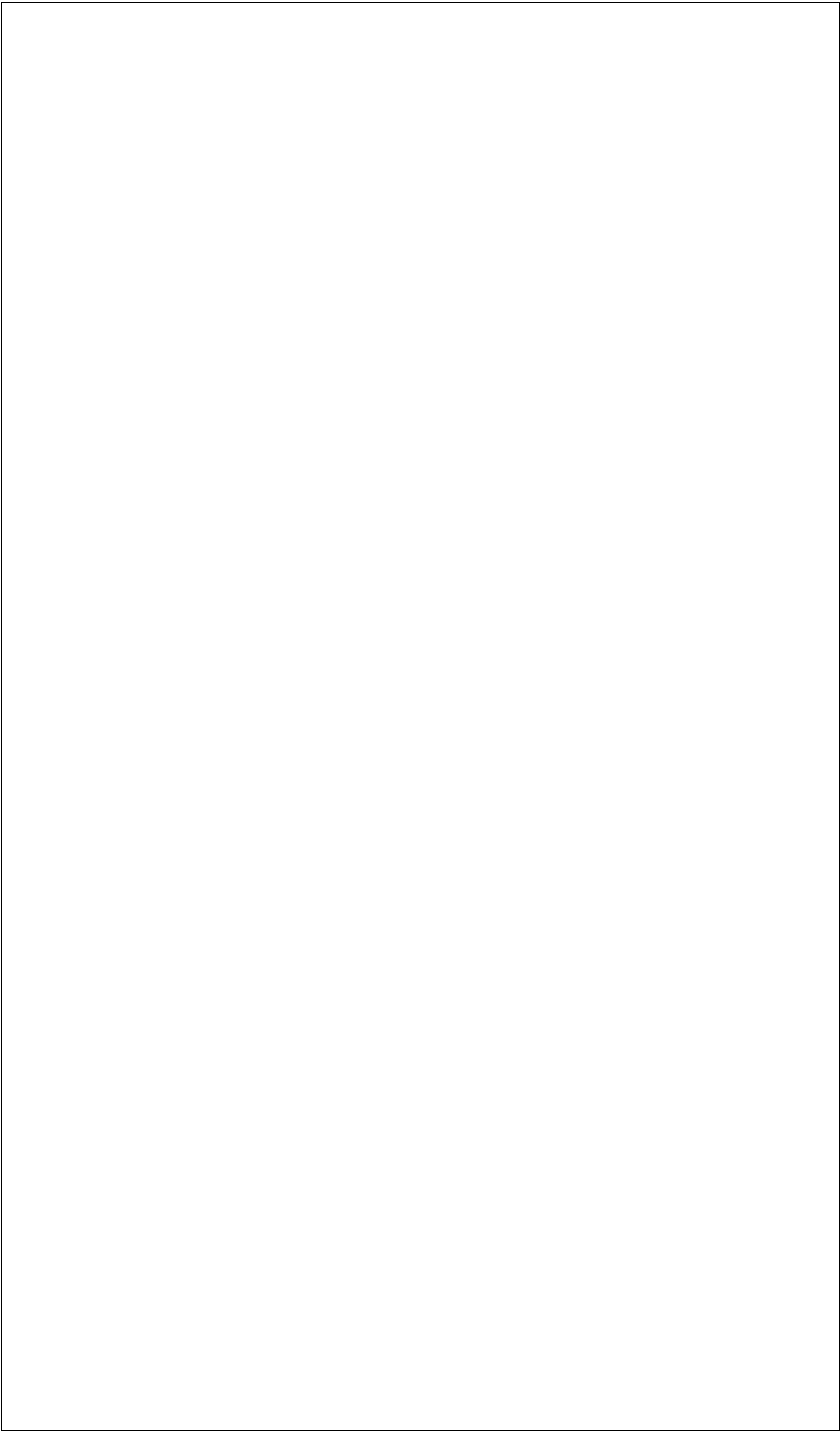
Poison Kill: Biometric-confirmed, irreversible connection termination. Zero-fills both devices. Server blacklists Connection ID.

Peer-Assisted Recovery: Phone lost? Same SIM on a new phone triggers P2P reconstruction of chat history from contacts.

Legal Shield Architecture: Metadata logs satisfy DoT retention rules. Zero knowledge of content. Provably compliant.

2.3 Operating Environment

Role	Access Level
Mobile	Android 14+ (API 34+), iOS 18+
HW requirements	Device with physical SIM slot or eSIM, biometric sensor camera for QR scanning
Network	4G/5G/Wi-Fi; active cellular connection required for initial SIM binding
Web	Chrome 120+, Firefox 120+, Safari 17+ (6-hour session limit enforced server-side)
Server	Linux-based cloud infrastructure (AWS/GCP), minimum 99.9% uptime SLA



2.4 User Classes and Characteristics

User Class	Profile	Technical Skill	Primary Features Used
Standard Ghost User	Privacy-conscious young professionals (18–35), urban, city-based	Medium-High	Daily QR, P2P messaging, Poison Kill
Service Provider	Lawyer, doctor, architect, CA seeking clients	Medium	Service ID QR, Verified badge, Tunnels
Group Admin	Team lead, community organizer, professional group owner	Medium-High	Group creation, Deputy assignment, GSK management
Service Taker	Client generating Service ID QR to find a professional	Low-Medium	Service ID QR generation, Tunnel management
System Administrator	SHRAVU internal staff	High	Compliance logs, server monitoring (no message access)

2.5 Design and Implementation Constraints

- DoT 2026 Mandate:** SIM binding MUST be hardware-verified (not OTP-only). Non-negotiable.
- 180-Day Metadata Retention:** Connection event logs, phone-ID mapping, and timestamps MUST be retained for a minimum of 180 days.
- No Cloud Chat Backup:** Chat history MUST NOT be backed up to any cloud server. Local-only.
- No Custom Cryptography:** Only use vetted standards — AES-256-GCM, Curve25519, HKDF-SHA256, ChaCha20-Poly1305.
- Zero-Fill Standard:** Poison Kill zero-fill MUST comply with DoD 5220.22-M standard for secure data erasure.
- Single Device Per SIM:** One active SHRAVU account per SIM. Multi-device login is prohibited by design.

2.6 Assumptions and Dependencies

- All users possess KYC-verified mobile numbers registered in India.
- Target devices support hw-level SIM API access (Andro Telephony Manager/iOS Core Telephony).
- Device system clock accuracy within ±30 seconds for QR nonce generation.
- Users access the internet at least once every 30 days to receive pending relay blobs.
- Push notification services (FCM for Android, APNs for iOS) are available and functional.
- Indian telecom operators expose IMSI verification APIs as required under DoT mandate.

SYSTEM FEATURES AND REQUIREMENTS

3.1 External Interface Requirements

3.1.1 User Interfaces (UI)

Screen	Description
Splash / Registration	Phone number entry → OTP → SIM binding progress → ID creation
Identity Dashboard	Displays SHRAVU ID, connection count, active tunnels
Daily QR Screen	Animated rotating QR with countdown timer showing expiry
Service ID QR Gen.	Form to set time limit (days), connection limit (count), optional message
Tunnel Chat View	Messages with P2P/Relay status indicator, media preview, Poison Kill button
Group Screen	Member list, admin controls, GSK epoch indicator, relay priority status
Poison Kill Confirmation	High-friction screen with red warning, biometric prompt, irreversibility warning
Recovery Screen	“Rebuilding from your trusted circle” progress with contact reconstruction status
Settings / Compliance	Subscription tier, audit log download, legal data request portal

3.1.2 Hardware Interfaces

Component	Description
SIM Card / eSIM	Android TelephonyManager.getSubscriberId() / iOS CoreTelephony (CTCarrier) to read IMSI + ICCID
Biometric Sensor	Android Biometric Prompt API / iOS Local Authentication framework for Poison Kill and app lock
Camera	System camera via QR scanning libraries (ZXing for Android, AV Foundation for iOS)
Secure Storage	Android Keystore / iOS Secure Enclave for private key storage

3.1.3 Software Interfaces

Interfaces	Description
Telecom Registry API	DoT-mandated endpoint for SIM binding cross-verification
FCM (Firebase Cloud Messaging)	Push notifications for message waiting signals (no payload)
APNS (Apple Push Noti. Service)	iOS push notifications
Payment Gateway	Razorpay for subscription billing (INR)
Biometric Authentication	OS-level biometric APIs (no third-party SDKs)

3.1.3 Communication Interfaces

Interfaces	Description
Protocol	WebSocket Secure (WSS) over TLS 1.3 for relay signalling
REST API	HTTPS over TLS 1.3 for registration, metadata logs, subscription management
P2P Comm.	WebRTC Data Channel for direct device-to-device messaging
Port	443 (standard HTTPS/WSS)
Certificate Pinning	SHA-256 certificate pinning enforced on all API endpoints

3.2 Use Case Specifications

PACKAGE 1 – IDENTITY & REGISTRATION

ID	Name	Pri. Actor	Description	Relationship
UC-1.1	Verify Phone Number	New User	User downloads the app and enters a 10-digit Indian mobile number. The system validates the format and sends an OTP via SMS within 30 seconds.	Extends UC-1.2
UC-1.2	Validate OTP	New User	User enters the 6-digit OTP received via SMS. The system verifies the OTP against the generated hash. The OTP expires after 5 minutes. The user has 3 attempts before a 15-minute lockout.	Include UC-1.3
UC-1.3	Perform SIM Binding Check	System	System queries device hardware SIM slot (IMSI + IMEI) to verify physical SIM presence. It cross-checks with the DoT telecom registry and confirms that the SIM is KYC-verified and active.	Include UC-1.4
UC-1.4	Create SHRAVU ID	System	System generates a unique @name#number ID. The user selects a preferred name component, and the system appends a unique numeric suffix. The ID is stored in admin logs linked to the phone number, which is hidden from other users.	Include UC-1.5
UC-1.5	Set Up Profile	New User	User sets an optional display name and avatar, selects notification preferences, and completes the onboarding tutorial covering Ghost mode, QR connections, and Poison Kill concepts.	Extends UC-1.4

PACKAGE 2 – CONNECTION MANAGEMENT

ID	Name	Pri. Actor	Description	Relationship
UC-2.1	Generate Daily Connection QR	Registered User	User taps "Connect" → "Daily QR". System generates QR tied to a Connection ID derived from SHRAVU ID + date seed + cryptographic nonce. QR displays with 24-hour countdown. QR is invalidated immediately upon scanning.	Include UC – 2.2
UC-2.2	Scan Connection QR	Connecting User	User Points Camera at Daily Connection QR. System decodes QR, validates Connection ID against server (checks not burned/expired), and sends connection request to QR owners. QR immediately burned on scan	Include UC – 2.3
UC-2.3	Accept / Reject Connection Request	QR Owner	QR Owner receives notification of connection request. Displays scanner’s SHRAVU ID. Owner taps Accept or Reject. On Accept: Cryptographic handshake occurs - both devices exchange public keys, bilateral tunnel established. On Reject: request deleted, no record.	Extends UC - 2.2
UC-2.4	Generate Service ID QR	Service Taker	User taps "Service QR". Sets Time Limit (default 30 days, range 1-365), Connection Limit (default 1, range 1-50), optional service description. System generates Service ID QR. User shares QR via any channel (social media, website, etc.).	Include UC-2.5, Include UC-2.6
UC-2.5	Claim Service ID QR	Service Provider	Provider scans Service ID QR. System validates: QR active, limit not reached, not expired. If valid: connection request sent to Service Taker with provider's Shravu ID. Service Taker must accept. On accept: Service Tunnel opened with defined time limit.	Include UC-2.6
UC-2.6	Service Tunnel Auto-Collapse	System	System monitors active Service Tunnels. At expiry timestamp: server sends Tunnel Collapse signal to both devices. Both devices lock the Service Tunnel chat. Local data retained unless Poison Kill invoked separately. User may request upgrade to permanent Ghost Handshake before expiry.	Extends UC-2.5

PACKAGE 3 – MESSAGING

ID	Name	Pri. Actor	Description	Relationship
UC-3.1	Send P2P Message (Online)	Sender	Sender types of messages in Tunnel view. App encrypts message with receiver's public key using AES-256-GCM. Encrypted payload sent directly via WebRTC Data Channel to receiver's active device. Receiver device decrypts with private key. Delivery confirmation returned.	None
UC-3.2	Sender Message via Relay (Offline)	Sender	Sender sends message; receiver is offline. App detects receiver offline via signalling server. Message encrypted with receiver’s public key. Encrypted blob sent to Relay Server with metadata (From Connection ID To Connection ID, expiry: 30 days for text). Server stores opaque blob, cannot decrypt.	Include UC – 3.3
UC-3.3	Relay Blob Delivery & Wipe	System	Receiver device comes online. Signaling server detects presence. Pushes notification to receiver device. Device retrieves encrypted blob from relay server. Device decrypts locally. Relay server immediately wipes the blob upon delivery acknowledgment (before decryption confirmation).	Extends UC-3.2
UC-3.4	Send Media File	Sender	Sender selects media file (image/video/document). App compresses and encrypts file. If receiver online: P2P transfer. If receiver offline: encrypted blob stored on relay server with 48-hour expiry (shorter than text). File size limits apply per tier.	Extends UC-3.1 or UC-3.2

PACKAGE 4 – GROUP MANAGEMENT

ID	Name	Pri. Actor	Description	Relationship
UC-4.1	Create Group	Group Admin	Admin taps "New Group". Enters group name, sets Member Limit (min 2), optionally pre-assigns Deputies (up to 3 from existing personal tunnels). System generates unique Group ID independent of all personal IDs. Group initially has 1 member (admin). No GSK yet.	Include UC-4.2
UC-4.2	Generate Group QR	Group Admin	System generates semi-permanent Group QR linked to Group ID with member limit embedded. Admin can share QR via any channel. Admin can revoke QR at any time — existing members unaffected, new joins use new QR.	Include UC-4.3
UC-4.3	Join Group via QR	New Group Member	User scans Group QR. System validates QR, member limit, and membership. Adds user to group. First member triggers GSK-001 creation by Admin via P2P; subsequent members trigger GSK rotation with new Salt and direct GSK delivery.	Include UC-5.1 or UC-5.2
UC-4.4	Revoke & Regenerate Group QR	Group Admin	Admin taps "Revoke QR" in group settings. Current Group QR immediately invalidated on server. New Group QR generated for the same Group ID. Existing members unaffected their membership tied to Group ID, not QR. New invitations must use new QR.	None
UC-4.5	Send Group Message	Group Member	Member sends group message. It is encrypted with current GSK (Layer 2) and for DRN (Layer 1). DRN delivers to online members; offline messages are queued (up to 6 hours). Message is tagged with current GSK epoch.	None
UC-4.6	In-Group Personal Connection Request	Group Member	Member A taps another Member B's name within group. Sends personal Connection Request — no QR exchange needed (group membership establishes trust layer). Member B receives notification and can Accept or Reject. On Accept: independent bilateral P2P tunnel opened between A and B, separate from group.	Extends UC-2.3
UC-4.7	Member Leaves Group	Group Member	Member taps "Leave Group". 10-min grace starts; Admin gets silent alert. After 10 min messages are zero-filled on all devices, kill signal queued (90-day expiry), and local copy erased. GSK rotates immediately; personal tunnels unaffected.	Include UC-5.2
UC-4.8	Admin Transfers Admin Role & Leaves	Group Admin	Admin must assign a new admin before leaving. After selection and confirmation, original admin's texts are zero-filled, kill signal dispatched GSK rotation occurs, and personal tunnels remain unaffected.	Include UC-4.7, Include UC-5.2

PACKAGE 5 – GSK CRYPTOGRAPHIC OPERATIONS

ID	Name	Pri. Actor	Description	Relationship
UC-5.1	GSK Initial Generation	System (Admin Device)	Triggered when first member joins a new group. Admin's device generates GSK-001 using CSPRNG (Cryptographically Secure Pseudo-Random Number Generator). GSK distributed to all current members (admin + first joiner) via their individual P2P tunnels. Group chat opens. Epoch counter set to 1.	Triggered by UC-4.3
UC-5.2	GSK Rotation on Membership Change	System (Admin / Deputy Device)	Any membership change triggers rotation. Admin device (or Deputy if admin offline — alphabetical fallback if deputies unavailable) generates 32-byte random Salt using CSPRNG. Salt is broadcast to all current members. Each existing member independently computes: NEW_GSK = HKDF (OLD_GSK, Salt, "SHRAVU -group-ratchet"). New joiners receive NEW_GSK directly via P2P tunnel. New epoch started. All messages < 30 days re-encrypted with NEW_GSK in background on each device.	Triggered by UC-4.3, UC-4.7, UC-4.8.
UC-5.3	GSK Burst Debounce	System	Multiple members joining within 2-second window triggers burst debounce. System queues all join events. After 2-second pause, processes all queued joins as a single batch event. ONE Salt generated. ONE new GSK computed. All batch-joiners receive the same new GSK. Single epoch increment.	Extends UC-5.2.
UC-5.4	GSK Distribution via P2P Tunnel	System	New GSK (or GSK for new joiners) distributed exclusively via each member's individual SIM-verified P2P tunnel — never through the group broadcast channel. Each member receives an individually encrypted copy. Distribution locked (group paused) until all current members acknowledge receipt. Lock released after all acknowledgments or 10-second timeout.	Extends UC-5.2.
UC-5.5	Old Message Re-Keying	System (On-Device)	After GSK rotation, background process re-encrypts all messages younger than 30 days with new GSK. Process runs silently during device idle time. Messages older than 30 days receive epoch tag for archive key lookup. Device maintains a maximum of 2 active GSK keys: current GSK + one archive GSK. Archive GSK auto-deleted when its epoch's messages expire or are deleted.	Extends UC-5.2.

PACKAGE 6 – POISON KILL

ID	Name	Pri. Actor	Description	Relationship
UC-6.1	Initiate Poison Kill	Initiator	User long-presses contact in tunnel list or navigates to tunnel settings → "Delete ID & Connection". App presents high-friction warning screen (red UI bold "IRREVERSIBLE" warning, explanation of consequences). User reads warning and taps - Proceed to Confirm	Includes UC-6.2.
UC-6.2	Biometric Confirmation	Initiator	System triggers biometric prompt (Face ID or Fingerprint) and verifies match. On match: Poison Kill executes. On failure: up to 3 attempts, then 5-minute lockout. No PIN fallback allowed — biometric only.	Includes UC-6.3 or UC-6.4.
UC-6.3	Kill Signal Delivery (Online)	System	System blacklists Connection ID on server and dispatches Kill Signal. Receiver locks chat and zero-fills local storage (DOD 5220.22-M); initiator simultaneously does the same. Both confirm erasure.	Extends UC-6.2.
UC-6.4	Kill Signal Delivery (Offline - Queued)	System	Receiver offline during Poison Kill. Kill Signal stored on server as metadata-only queue item. On next SHRAVU open, it is delivered before any action; chat locks and storage are zero-filled. 90-day expiry—if not reconnected, signal expires, local data remains but tunnel is permanently blacklisted.	Extends UC-6.2.

PACKAGE 7 – RECOVERY

ID	Name	Pri. Actor	Description	Relationship
UC-7.1	Detect New Device with Same SIM	System	User installs SHRAVU on a new device with the same SIM. OTP → SIM binding. Server detects previous SIM on different hardware, confirms new device (new IMEI, same IMSI) as "Recovery Registration," and pings all active Connection IDs for this SHRAVU ID.	Includes UC-7.2.
UC-7.2	Peer History Reconstruction	System / Contact Devices	Each pinged contact recognizes the Recovery Ping and auto-initiates P2P transfer of shared chat history to the new device. No consent needed (bilateral ownership). Process continues until all responsive contacts send their copies.	Extends UC-7.1.
UC-7.3	Verify Recovery Completeness	Recovering User	App displays Recovery Summary: "X contacts responded, Y tunnels restored, Z tunnels unavailable." Unavailable tunnels listed as: Poison-Killed (permanently gone), Contact.	Extends UC-7.2.

PACKAGE 8 – COMPLIANCE & ADMIN

ID	Name	Pri. Actor	Description	Relationship
UC-8.1	Generate Metadata Log for Authority Request	System Administrator	Compliance officer receives legal notice, accesses portal, and queries by phone or SHRAVU ID. System generates metadata report (phone↔ID, connection events, SIM bindings, session logs) with no message content, exported as signed PDF/JSON, retained 180 days.	None
UC-8.2	Demonstrate Zero-Knowledge Compliance	System Administrator	For any authority request, system issues a Zero-Knowledge Compliance Certificate confirming: (1) message content was never stored, (2) Poison Kill zero-filled content on all devices, (3) recovery is impossible. Certificate is cryptographically signed by SHRAVU’s legal key.	None
UC-8.3	Web Session Authentication & 6-Hour Logout	Web User	User accesses SHRAVU Web via SIM-verified QR scan. Session runs 6 hours server-side, then ends; re-authentication via QR required. No "remember me" option (DoT mandate).	None

PACKAGE 9 – SUBSCRIPTION MANAGEMENT

ID	Name	Pri. Actor	Description	Relationship
UC-9.1	Upgrade to Pro Tunnel	Registered User	User goes to Settings → Subscription → "Upgrade to Pro", selects plan (₹99 monthly / ₹249 3-month / ₹449 6-month). Razorpay processes payment. On success, account upgrades instantly with expanded limits and features.	None
UC-9.2	Downgrade / Cancel Subscription	Subscribed User	User goes to Settings → Subscription → Cancel. System shows features lost. User confirms. At current billing period end, tier reverts to Free Ghost. Excess tunnels (above 10) and groups become read-only; no new messages. Data remains, only access restricted.	None

3.3 Functional Requirement

3.3.1 – IDENTITY & REGISTRATION – FR – I series

Req ID	Requirement Statement	Priority	Acceptance Criteria	Related UC
FR-I1.1	The system SHALL send OTP within 30 seconds of phone number submission.	CRITICAL	• OTP delivered within 30s in 99% of cases • 6-digit numeric OTP • Expires after 300 seconds	UC-1.1
FR-I1.2	The system SHALL lock account for 15 minutes after 3 consecutive failed OTP attempts.	HIGH	• Counter resets after successful login • Lockout message displayed	UC-1.2
FR-I1.3	The system SHALL perform hardware-level SIM binding using IMSI + IMEI cross-verification.	CRITICAL	• IMSI read from device via Telephony Manager / CoreTelephony • IMEI logged in compliance store • Cross-verified with DoT registry	UC-1.3
FR-I1.4	The system SHALL enforce one SHRAVU ID per phone number.	CRITICAL	• Duplicate registration attempt returns error • Previous session invalidated on new device binding	UC-1.3
FR-I1.5	The system SHALL create SHRAVU ID in format @name#uniquenumber.	HIGH	• Numeric suffix unique across platform • Name component set by user (3–20 characters)	UC-1.4
FR-I1.6	The system SHALL make phone number invisible to all users except SHRAVU admin logs.	CRITICAL	• No API endpoint returns phone number to other users • Phone visible only in compliance portal	UC-1.4
FR-I1.7	The system SHALL enable Ghost mode by default (no discovery, no contact sync).	CRITICAL	• No global search function • No suggested contacts • User discoverable only via QR scan	UC-1.4
FR-I1.8	The system SHALL enforce automatic web session logout after exactly 6 hours.	CRITICAL	• Server-side timer, cannot be extended client-side • New QR scan required to resume	UC-8.3

3.3.2 – CONNECTION MANAGEMENT – FR – C series

Req ID	Requirement Statement	Priority	Acceptance Criteria	Related UC
FR-C1.1	The system SHALL generate a Daily Connection QR tied to a Connection ID, rotating every 24 hours.	CRITICAL	• New Connection ID generated each UTC Day • Previous day's ID expires at midnight	UC-2.1
FR-C1.2	The system SHALL burn (invalidate) the Daily QR immediately upon first scan.	CRITICAL	• QR invalidated within 200ms of scan detection • Second scan of same QR returns "expired" error	UC-2.2
FR-C1.3	The system SHALL require explicit Accept confirmation from the QR owner before tunnel opens.	HIGH	• Tunnel NOT opened on scan alone • Accept button required from QR owner	UC-2.3
FR-C1.4	The system SHALL perform Curve25519 key exchange during tunnel establishment.	CRITICAL	• Key exchange completed within 3 seconds • Keys stored in device Secure Enclave/Keystore	UC-2.3
FR-C1.5	The system SHALL allow Service ID QR with configurable time limit (1–365 days) and connection limit (1–50).	HIGH	• Default: 30 days, limit 1 • Admin can modify before sharing	UC-2.4
FR-C1.6	The system SHALL auto-collapse Service Tunnel at expiry timestamp.	HIGH	• Collapse signal sent to both devices at expiry ± 60 seconds • Tunnel locked, no new messages	UC-2.6
FR-C1.7	The system SHALL allow Service Tunnel upgrade to permanent Ghost Handshake anytime before expiry.	MEDIUM	• Upgrade option visible in tunnel settings • Both parties must confirm upgrade	UC-2.6
FR-C1.8	The system SHALL prevent any user from finding another user via global search or phone number lookup.	CRITICAL	• No search API endpoint for users • Phone-to-ID lookup blocked for all non-admin roles	UC-1.4

3.3.3 – COMMUNICATION & RELAY – FR – M series

Req ID	Requirement Statement	Priority	Acceptance Criteria	Related UC
FR-M1.1	The system SHALL encrypt all P2P messages using AES-256-GCM with Curve25519 key agreement.	CRITICAL	• LibSodium or equivalent vetted library • No message content stored on server	UC-3.1
FR-M1.2	The system SHALL route messages directly P2P when both devices are online (WebRTC Data Channel).	HIGH	• P2P latency < 300ms • Server sees only routing metadata	UC-3.1
FR-M1.3	The system SHALL store encrypted relay blobs for offline receivers with text expiry of 30 days.	CRITICAL	• Blob encrypted before leaving sender device • Server cannot decrypt blob	UC-3.2
FR-M1.4	The system SHALL store encrypted media relay blobs with 48-hour expiry.	CRITICAL	• Media size limits up to 200MB • Blob wiped after 48 hours	UC-3.4
FR-M1.5	The system SHALL immediately wipe relay blob from server upon delivery confirmation.	CRITICAL	• Wipe executes before decryption confirmation • No server copies after delivery	UC-3.3
FR-M1.6	The system SHALL send push notification (FCM/APNS) with NO message content when relay blob arrives.	HIGH	• Notification payload: sender ID only • No message preview in notification	UC-3.3
FR-M1.7	The system SHALL display P2P/Relay status indicator in chat UI.	MEDIUM	• Green indicator for P2P • Yellow indicator for relay • Gray for undelivered	UC-3.1
FR-M1.8	The system SHALL enforce certificate pinning on all server communications.	CRITICAL	• SHA-256 pin validation • Connection rejected on pin mismatch	UC-3.1

3.3.4 – GROUP MANAGEMENT – FR – G series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-G1.1	The system SHALL assign a unique Group ID to each group, independent of all personal IDs.	CRITICAL	- Group ID format: grp - #UUID - Not derived from admin’s SHRAVU ID	UC-4.1
FR-G1.2	The system SHALL allow admin to set a member limit (minimum 2) at group creation.	HIGH	- Default limits: Free = 20, Pro = 100, Verified = 500 - Limit cannot be set below current member count	UC-4.1
FR-G1.3	The system SHALL open group chat immediately when the first non-admin member joins.	CRITICAL	- Chat accessible with 2 members (admin + 1) - No minimum quorum required before messaging	UC-4.3
FR-G1.4	The system SHALL automatically rotate the GSK on every group membership change event.	CRITICAL	- Triggers: join, leave, admin change, limit change - Rotation completes within 5 sec - No user action required	UC-5.2
FR-G1.5	The system SHALL ensure new joiners cannot access or decrypt messages sent before their epoch.	CRITICAL	- New member receives current GSK only, never historical GSKs - Historical message content cryptographically inaccessible to late joiners	UC-4.3
FR-G1.6	The system SHALL implement a Priority Relay Stack for group message routing.	HIGH	- Stack order: Admin → Deputy 1 → Deputy 2 → Alphabetical online member → Server relay - Failover within 3 seconds - Server relay max 6 hrs for groups	UC-4.5
FR-G1.7	The system SHALL allow admin to revoke and regenerate Group QR without disbanding the group or removing existing members.	HIGH	- Revoke + regenerate in single action - Existing members' membership unaffected - New QR immediately valid after generation	UC-4.4
FR-G1.8	The system SHALL enforce a 10-minute grace window before executing a departing member's message erasure.	HIGH	- Silent metadata alert sent to admin on departure initiation - Erasure begins after exactly 600 seconds - Admin can flag conversation during this window	UC-4.7
FR-G1.9	The system SHALL allow members to send personal connection requests to other members within the same group without requiring QR exchange.	MEDIUM	- Connection request initiated via profile tap in group member list - Resulting bilateral tunnel is independent of the group - Tunnel survives group exit by either member	UC-4.6
FR-G1.10	The system SHALL prevent the admin from leaving a group without first designating a new admin.	CRITICAL	- Leave Group button disabled until new admin is selected and confirmed - System prompt displayed on leave attempt - New admin assignment is irreversible once confirmed	UC-4.8

3.3.5 – GSK CRYPTOGRAPHIC ENGINE – FR – K series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-K1.1	The system SHALL generate the initial GSK using a CSPRNG with minimum 256-bit entropy.	CRITICAL	• CSPRNG: /dev/random on Android, SecRandomCopyBytes on iOS • GSK-001 created when first member joins • Entropy verified before key use	UC-5.1
FR-K1.2	The system SHALL implement GSK rotation using the formula: NEW_GSK = HKDF-SHA256(OLD_GSK, Salt, "shravu-group-ratchet").	CRITICAL	• Salt: 32-byte random value • KDF: IETF RFC 5869 HKDF-SHA256 • Context string: "shravu-group-ratchet" hardcoded	UC-5.2
FR-K1.3	The system SHALL broadcast only the 32-byte Salt to existing members during GSK rotation, never the full new GSK value.	CRITICAL	• Server receives and relays only Salt (32 bytes) • Full new GSK computed locally on each member device • New joiners receive full GSK via direct P2P tunnel only	UC-5.2
FR-K1.4	The system SHALL implement a 2-second burst debounce for simultaneous join events.	HIGH	• Multiple joins within 2-second window treated as a single batch • Only one GSK rotation event triggered per batch • All batch members receive same new GSK epoch	UC-5.3
FR-K1.5	The system SHALL implement a GSK distribution lock protocol that pauses group messaging during key distribution.	CRITICAL	• Group message sending disabled during distribution • Lock released after all members acknowledge or 10-second timeout • Queued messages delivered after lock release	UC-5.4
FR-K1.6	The system SHALL maintain a maximum of 2 active GSK keys per device at any time: the current GSK and one archive GSK.	HIGH	• Archive GSK auto-deleted when its epoch's messages are confirmed delivered or expired • No unbounded accumulation of historical keys	UC-5.5
FR-K1.7	The system SHALL re-encrypt messages younger than 30 days with the new GSK in a background process after rotation.	HIGH	• Re-encryption runs during device idle state • No UI blocking or user action required • Process restartable if interrupted	UC-5.5
FR-K1.8	The system SHALL cryptographically orphan a departed member's GSK, making it impossible to derive future keys.	CRITICAL	• Ex-member holds old GSK but has no access to new Salt • Cannot compute new GSK without Salt • Mathematically verified: HKDF without Salt yields indistinguishable output	UC-5.2

3.3.6 – POISON KILL SYSTEM – FR – P series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-P1.1	The system SHALL require biometric authentication (Face ID or Fingerprint only) to execute a Poison Kill.	CRITICAL	• No PIN or password fallback accepted for Poison Kill • Maximum 3 biometric attempts • 5-minute lockout after 3 failed attempts	UC-6.2
FR-P1.2	The system SHALL permanently and irreversibly blacklist the killed Connection ID on the server.	CRITICAL	• Blacklist entry is permanent, no expiry • Blacklisted ID cannot be reused or re-activated • Blacklist checked on all inbound connection attempts	UC-6.3
FR-P1.3	The system SHALL zero-fill all local storage for the killed connection to DOD 5220.22-M standard on both devices.	CRITICAL	• All messages, media, keys, and metadata overwritten with binary zeros • Multi-pass overwrites as per DOD standard • Verified by checksum comparison post-wipe	UC-6.3
FR-P1.4	The system SHALL dispatch a Kill Signal to an online receiver device within 2 seconds of Poison Kill execution.	HIGH	• Kill Signal delivery within 2 seconds (online) in 99% of cases • Chat interface locks on receiver side within 3 seconds of signal receipt	UC-6.3
FR-P1.5	The system SHALL queue the Kill Signal for offline receivers with a 90-day expiry window.	HIGH	• Queue item contains: Connection ID + kill timestamp only (no content) • Kill Signal delivered immediately on receiver's next login • Queue item auto-expires and is deleted at 90 days	UC-6.4
FR-P1.6	The system SHALL display a high-friction irreversibility warning screen before biometric confirmation.	MEDIUM	• Red UI theme on warning screen • Text "IRREVERSIBLE ACTION" displayed prominently • Minimum 3-second forced delay before confirm button activates	UC-6.1

3.3.7 – PEER-ASSISTED RECOVERY – FR – R series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-R1.1	The system SHALL detect a new device registration using a previously bound SIM and automatically initiate the recovery protocol.	CRITICAL	• New IMEI detected with existing IMSI → recovery flow triggered • No user action required beyond SIM binding on new device	UC-7.1
FR-R1.2	The system SHALL ping all active Connection IDs associated with the recovering Shravu ID.	CRITICAL	• Pings sent via signaling server using metadata only • No message content transmitted in ping • Pings sent within 60 seconds of recovery detection	UC-7.2
FR-R1.3	The system SHALL allow contact devices to automatically transmit shared chat history to the recovering device via P2P tunnel without requiring explicit consent.	HIGH	• Transmission via SIM-verified P2P tunnel only • History is bilateral — no consent required as both parties own it • Auto-transmission begins within 30 seconds of recovery ping receipt	UC-7.2
FR-R1.4	The system SHALL permanently prohibit recovery of any Poison-Killed tunnel under any circumstance.	CRITICAL	• Killed Connection IDs permanently blacklisted, no recovery path exists • Recovery ping NOT sent for killed Connection IDs • This behaviour documented in user-facing terms of service	UC-7.2
FR-R1.5	The system SHALL display a recovery completeness summary to the recovering user upon completion.	MEDIUM	• Summary shows: tunnels restored (count), tunnels unavailable (count + reason) • Reasons listed: "Poison-Killed", "Contact Offline", "Contact Uninstalled"	UC-7.3

3.3.8 – LEEGAL COMPLIANCE & ADMIN LAYER– FR – L series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-L1.1	The system SHALL retain Phone↔ID mapping, connection event logs, and timestamps for a minimum of 180 days.	CRITICAL	• Retained in compliance-grade PostgreSQL database • Encrypted at rest with AES-256 • Automated retention policy enforced with deletion logs	UC-8.1
FR-L1.2	The system SHALL generate a metadata compliance report within 24 hours of a valid authority request.	HIGH	• Report includes: Phone No, Shravu ID, connection events, timestamps, SIM binding records, web session logs • Report exported as signed PDF and JSON • Contains NO message content	UC-8.1
FR-L1.3	The system SHALL generate a cryptographically signed Zero-Knowledge Compliance Certificate when message content is requested but architecturally unavailable.	HIGH	• Certificate states architectural impossibility of content retrieval • Signed by Shravu's legal infrastructure key pair • Certificate includes technical architecture attestation	UC-8.2
FR-L1.4	The system SHALL maintain an immutable audit trail of all compliance portal access events.	HIGH	• Logs: accessor identity, timestamp, query parameters, report generated • Audit trail itself encrypted and tamper-evident • Retained for 180 days minimum	UC-8.1
FR-L1.5	The system SHALL store metadata logs on a dedicated compliance server cluster segregated from the relay infrastructure.	HIGH	• Separate database cluster with separate access credentials • No shared admin access between relay and compliance clusters	UC-8.1
FR-L1.6	The system SHALL provide a grievance officer contact mechanism as required by the IT Act 2000.	MEDIUM	• In-app link to grievance portal prominently displayed in Settings • Response acknowledgment within 24 hours, resolution within 72 hours per Act • Grievance officer name and contact details publicly disclosed	UC-8.1

3.3.9 – SUBSCRIPTION & BILLING – FR – S series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-S1.1	The system SHALL enforce Free Ghost tier limits: 10 tunnels, 1 group (max 20 members), 1 active Service ID QR.	HIGH	• Tunnel creation blocked at limit with clear UI message • Soft limit displayed as counter (e.g., "9/10 tunnels used")	UC-9.1
FR-S1.2	The system SHALL process subscription payments via Razorpay with INR pricing across monthly, 3-month, and 6-month billing cycles.	HIGH	• Pro: ₹99/month, ₹249/3mo, ₹449/6mo • Verified: ₹299/month, ₹749/3mo, ₹1399/6mo • Auto-renewal with 3-day advance notification	UC-9.1
FR-S1.3	The system SHALL require document-based verification for the Verified professional badge, with manual review completed within 48 hours.	HIGH	• Accepted documents: Bar Council ID, Medical Registration, CA/ICAI Certificate • Manual review by Shravu verification team • Rejection includes reason and re-submission option	UC-9.2
FR-S1.4	The system SHALL automatically remove the Verified badge after a 7-day grace period following subscription lapse.	HIGH	• 7-day warning notification before badge removal • Badge removed on day 8 after lapse if not renewed • Badge restoration requires re-verification	UC-9.3
FR-S1.5	The system SHALL make excess tunnels and groups read-only (not delete) upon subscription downgrade.	MEDIUM	• Existing data preserved in all tunnels and groups • New messages blocked in tunnels exceeding Free tier limit • Clear UI indicator: "Upgrade to send messages in this tunnel"	UC-9.4
FR-S1.6	The system SHALL provide an itemized GST-compliant invoice for all subscription transactions.	LOW	• Invoice includes amount, GST breakdown (18%), billing period, payment method • Downloadable as PDF from Settings → Billing History	UC-9.1

3.3.10 – Web Session Management – FR – W series

Req ID	Req Statement	Priority	Acceptance Criteria	Rel. UC
FR-W1.1	The system SHALL require a SIM-verified QR scan from the primary phone to initiate any web session.	CRITICAL	• Web session QR expires 60 seconds after generation on phone • New QR required for each new web session initiation • Session starts only after QR verified against live SIM binding	UC-8.3
FR-W1.2	The system SHALL forcibly terminate web sessions after exactly 6 hours regardless of user activity.	CRITICAL	• 6-hour timer enforced server-side, not client-side • Session terminated even during active use • User notified 5 minutes before session expiry • Hardcoded per DoT 2026 mandate, non-configurable	UC-8.3
FR-W1.3	The system SHALL NOT offer any "Remember Me", persistent session, or session extension option for web access.	CRITICAL	• No session cookies persisted beyond 6-hour limit • No local storage of session tokens in browser • Re-authentication via phone QR scan required on every session	UC-8.3
FR-W1.4	The system SHALL log all web session start and end events in the compliance logs.	HIGH	• Log entries include session start timestamp, session end timestamp, session duration, hashed IP address • Retained in compliance logs for 180 days minimum	UC-8.3

3.4 Non-Functional Requirement

3.4.1 – PERFORMANCE REQUIREMENTS (NFR-P)

Req ID	Requirement	Metric	Implementation	Rational
NFR-P1	P2P Message Delivery Latency	< 300 ms (95th percentile)	WebRTC Data Channel direct device-to-device connection	Real-time chat user experience; competitive parity with WhatsApp
NFR-P2	Relay Blob Push Notification delay	< 5 seconds after device comes online	FCM/APNS high-priority push channel; pre-positioned relay blob	User awareness of pending messages within seconds
NFR-P3	App Launch to Usable State	< 1.5 seconds on mid-range device (3GB RAM, Snapdragon 7 series)	Native code optimization; lazy loading of non-critical modules; splash screen preloading	User retention; India mid-range device market dominates
NFR-P4	GSK Rotation Completion Time	< 5 seconds for groups up to 100 members	Async parallel key distribution via pre-established P2P tunnels; burst frequency debounce reduces event	GSK rotation must be imperceptible to end users during group conversations
NFR-P5	QR Code Generation	< 200 ms from trigger to display	On-device QR generation (ZXing/CoreImage); pre-computed daily nonce library	Instant QR display essential for in-person handshake UX
NFR-P6	Poison Kill Zero-Fill Execution	< 10 seconds for up to 10,000 messages locally stored	DOD 5220.22-M block-level overwrite; background thread with progress indicator	Security guarantee must complete before user leaves the screen

3.4.2 – SECURITY REQUIREMENTS (NFR-S)

Req ID	Requirement	Metric	Implementation	Rational
NFR-S1	Message Encryption Standard	AES-256-GCM + Curve25519 ECDH key agreement	LibSodium library (audited, open source); hardware-backed keystore via OS Secure Enclave / Android Keystore	Industry-go-resistant to all known classical attacks
NFR-S2	Cryptographic Key Storage	100% hardware-backed; keys never exported to user space	Android Key Store System (StrongBox); iOS Secure Enclave; keys never leave hardware boundary	Prevents key extraction via memory dumps, rooted devices, or debugging attacks
NFR-S3	Transport Security	TLS 1.3 minimum; SHA-256 certificate pinning on all endpoints	Network Security Config (Android); NSAppTransportSecurity (iOS) pin rotation procedure documented	MITM attack prevention; protects all server communications including metadata
NFR-S4	Zero-Knowledge Server Architecture	0% message content readable by server at any time	Architecture-enforced: relay stores ciphertext only; no server-side decryption keys exist	Privacy-by-design; legally protects Shravu from content disclosure demands
NFR-S5	Secure Data Erasure Standard	DOD 5220.22-M compliant zero-fill for Poison Kill	Multi-pass binary zero overwrite at block level; post-wipe checksum verification	Ensures forensic irrecoverability; legally defensible erasure standard
NFR-S6	GSK Forward & Backward Secrecy	Compromise of one GSK epoch yields access to zero other epochs	HKDF one-way derivation ensures backward secrecy; Salt randomness ensures forward secrecy; max 2 keys per device	Perfect Forward Secrecy (PFS) + Post-Compromise Security (PCS) — Signal-grade standard
NFR-S7	Biometric Authentication Strength	FIDO2-compliant; liveness detection required for Face ID; Class 3 biometric for fingerprint	Android BiometricPrompt Class 3 (strong); iOS Face ID with liveness detection; no PIN fallback for Poison Kill	Anti-spoofing protection; prevents Poison Kill execution by unauthorized person
NFR-S8	Security Testing & Vulnerability Management	Annual external penetration test; continuous automated scanning; HackerOne bug bounty program	Engage certified pentesting firm annually; SAST/DAST in CI/CD pipeline; public bug bounty with defined severity rewards	Ongoing security assurance; builds user trust; industry standard for security-critical applications

3.4.3 – AVAILABILITY & RELIABILITY REQUIREMENTS (NFR-R)

Req ID	Requirement	Metric	Implementation	Rational
NFR-R1	Relay Server Uptime	99.9% (≤ 8.7 hours downtime per year)	Multi-region load balanced cluster across AWS Mumbai + Hyderabad; automated health checks every 30 seconds	Message delivery guarantee; relay downtime means undelivered messages for offline users
NFR-R2	Signalling Server Uptime	99.9% with active-active redundancy	Active-active redundancy across 2 geographic regions; sub-30-second failover via DNS health checks	P2P connection establishment depends on signalling server; downtime prevents new connections
NFR-R3	Compliance Server Uptime	99.95% (higher SLA than relay — legal obligation)	Dedicated compliance cluster entirely separate from relay infrastructure; 3x replication	Legal compliance data must be available 24/7; authority requests cannot wait for downtime windows
NFR-R4	Relay Blob Durability	99.999% (no data loss for stored relay blobs)	3x replication across availability zones; immediate replication before delivery confirmation	Undelivered messages are the user's only copy; loss is unacceptable
NFR-R5	Disaster Recovery RTO (Recovery Time Objective)	< 4 hours for full-service restoration	Automated failover procedures; documented runbooks; quarterly DR drills	Business continuity; meets reasonable user expectations for a messaging service

3.4.4 – SCALABILITY REQUIREMENTS (NFR-SC)

Req ID	Requirement	Metric	Implementation	Rational
NFR-SC1	Concurrent Active Tunnels	10 million tunnels at launch capacity	Horizontal relay server scaling; stateless relay architecture; sharding by Connection ID hash	Targeting 5 million users at launch, with an average of 2 tunnels per user
NFR-SC2	Peak Message Throughput	100,000 messages / second	Apache Kafka message queue for relay ingestion; distributed relay node processing; horizontal auto-scaling	Peak load events (festivals, breaking news, viral onboarding) must not degrade service.
NFR-SC3	Simultaneous GSK Rotation Events	1,000 concurrent group membership changes without degradation	Asynchronous GSK rotation queue; 2-second burst debounce reduces event frequency; distributed key distribution workers	Large community events (conference groups, college orientation) can trigger mass simultaneous joins.
NFR-SC4	New User Registration Rate	50,000 registrations per hour during launch campaigns	Auto-scaling registration service; SIM binding via DoT API with rate limiting and caching; OTP delivery through multiple SMS gateway providers	Marketing campaign bursts (Product Hunt launch, influencer onboarding) require elastic scaling.

3.4.5 – USABILITY REQUIREMENTS (NFR-U)

Req ID	Requirement	Metric	Implementation	Rational
NFR-U1	Onboarding completion rate.	Onboarding completion rate: > 80% of app installs complete registration through SIM binding.	Accessibility: Simplified 3-step onboarding flow; permission requests include clear rationale; tutorial is skip-able.	Industry average onboarding completion is 60–70%; 80% target reflects streamlined UX.
NFR-U2	First QR connection success rate.	First QR connection success rate: > 90% on first attempt.	Camera permission guided request: QR minimum size 200×200 px; contrast ratio guidelines followed for display; fallback manual code entry provided.	First connection experience is the critical activation moment; failure here equals churn.
NFR-U3	App accessibility.	App accessibility: WCAG 2.1 Level AA compliant.	Screen reader support (TalkBack / VoiceOver); minimum 4.5:1 color contrast ratio; touch target minimum 44×44 pt; font scaling supported.	Inclusive design; accessibility compliance increasingly required under India's disability rights frameworks.
NFR-U4	Language support at launch.	Language support at launch: Hindi and English (v2.0); Tamil, Telugu, Kannada, and Bengali targeted for v2.1.	i18n framework (react-native-i18n or flutter_intl); all strings externalized; RTL support included; architecture prepared for future Urdu support.	India’s diverse linguistic market; Hindi + English covers ~70% of target user base at launch.
NFR-U5	Error message clarity.	Error message clarity: 100% of error states have human-readable, actionable messages.	User research–validated error copy; no technical error codes shown to end users; every error includes “what to do next” guidance.	User trust and retention: technical jargon in errors causes abandonment, especially among Tier-2 city users.

APPENDICES

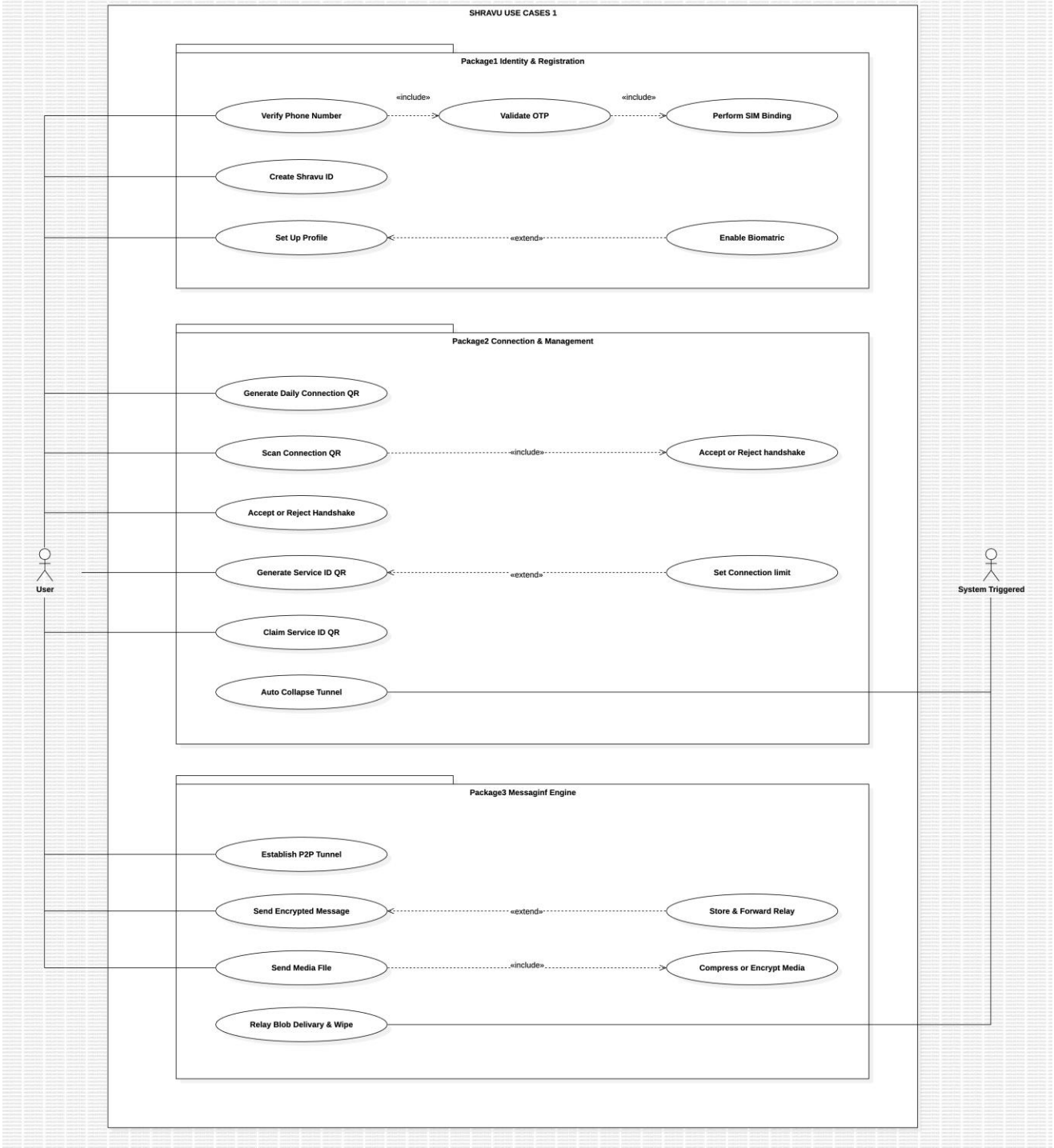
4.1 Glossary

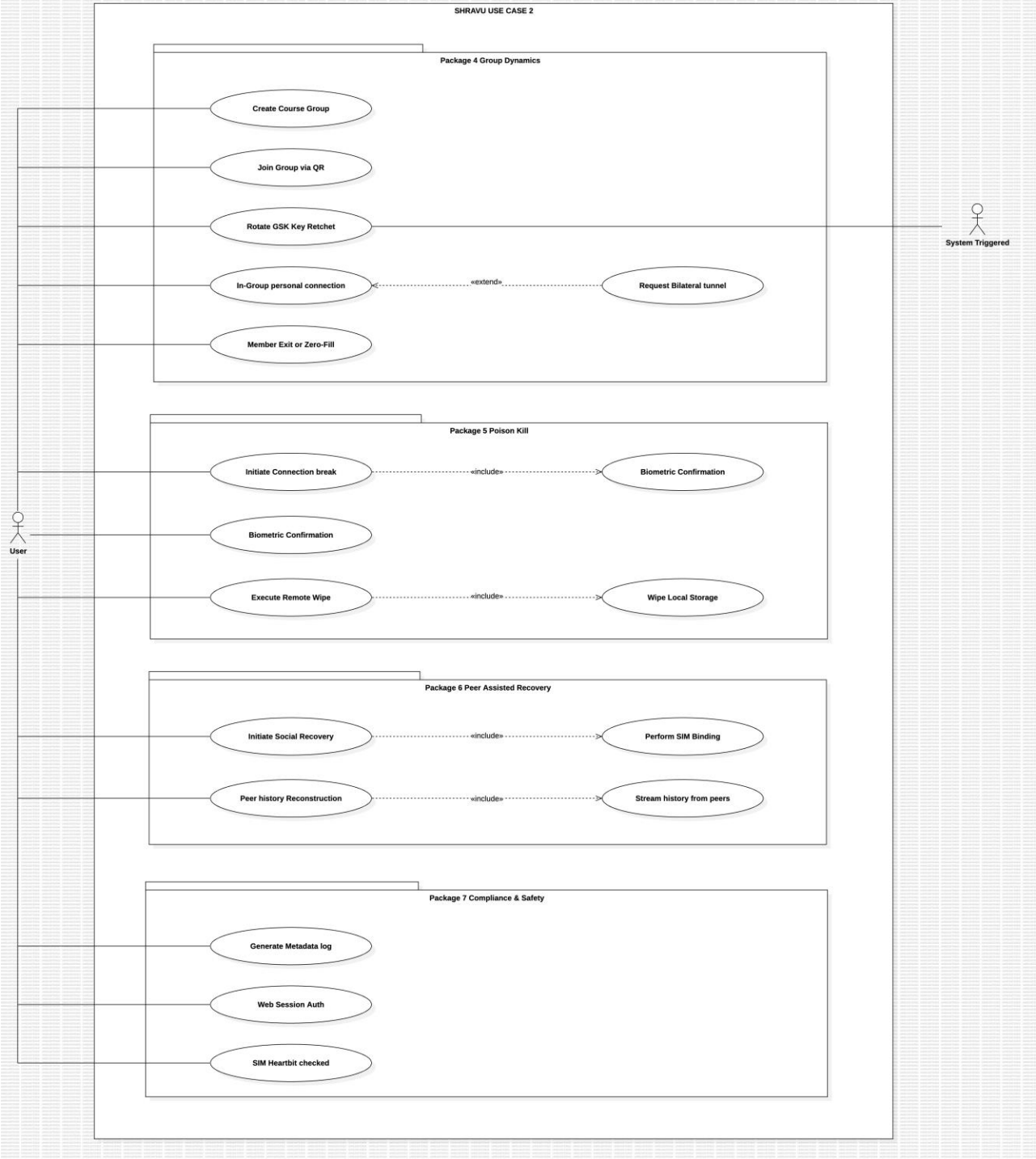
Term	Defination
AES – 256 -GCM	Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode. Authenticated encryption algorithm used for all Shravu message encryption. Provides both confidentiality and integrity verification.
Archive GSK	The single previous-epoch Group Session Key retained on a device for decrypting messages older than 30 days. Devices maintain a maximum of one archived GSK at any time alongside the current GSK.
Bilateral Tunnel	A P2P encrypted communication channel between exactly two Shravu IDs, established through the Ghost Handshake protocol. Each tunnel is cryptographically independent of all other tunnels.
Burst Debounce	A 2-second event coalescing window that combines multiple rapid group membership changes (joins within a 2-second window) into a single GSK rotation event, preventing key rotation thrashing during mass onboarding.
Connection ID	A temporary cryptographic identifier used during the Ghost Handshake. For personal connections, it rotates every 24 hours. For service connections, it is defined by the Service ID QR parameters. Burned (invalidated) upon first use.
CSPRNG	Cryptographically Secure Pseudo-Random Number Generator. Used for generating GSK material and salts. Implementation: /dev/urandom on Android, SecRandomCopyBytes on iOS.
Curve25519	An elliptic-curve Diffie-Hellman (ECDH) function used for key agreement during bilateral tunnel establishment. Generates a shared secret from which AES-256 session keys are derived.
DRN (Designated Relay Node)	The group member responsible for routing group messages to all other members. Priority order: Admin → Deputy 1 → Deputy 2 → alphabetical online member → server relay (6-hour max).
Epoch	A cryptographic period defined by a single GSK version. Each GSK rotation creates a new epoch with an incremented epoch number. All messages are tagged with their epoch for correct key lookup during decryption.
Ghost by Default	Shravu's core privacy principle: all users are invisible and undiscoverable by default. A user can only be found by another user if they actively share a QR code. No global search, no contact discovery, no suggested connections.
Ghost Handshake	The proprietary connection establishment protocol unique to Shravu. Involves generating an ephemeral, single-use QR code that is cryptographically burned on scan, followed by a bilateral key exchange and an explicit Accept step.
GSK (Group Session key)	A shared symmetric encryption key held by all current members of a Shravu group. Used to encrypt group messages for all members (Layer 2). Rotates automatically on every membership change event.
HKDF	HMAC-based Key Derivation Function as defined in IETF RFC 5869. Used for GSK ratchet: NEW_GSK = HKDF-SHA256(OLD_GSK, Salt, "shravu-group-ratchet"). One-way function — cannot be reversed to recover OLD_GSK.
IMEI	International Mobile Equipment Identity (IMEI). A unique 15-digit hardware identifier embedded in every mobile device. Used in SIM binding to link the account to a specific physical device.

Term	Defination
IMSI	International Mobile Subscriber Identity (IMSI). A unique identifier embedded in the SIM card. Used in SIM binding to identify the KYC-verified subscriber associated with the account.
KDF	Key Derivation Function. Any one-way mathematical function that derives a new cryptographic key from existing key material. Shravu uses HKDF-SHA256 as its KDF for the GSK ratchet chain.
Poison kill	Shravu's irreversible connection termination protocol. Requires biometric confirmation. Execution: (1) Connection ID permanently blacklisted on the server. (2) Kill Signal dispatched to both devices. (3) All local storage for that connection zero-filled on both devices. Irreversible by design.
Relay Blob	An opaque, encrypted message payload stored temporarily on Shravu's relay server when the intended recipient device is offline. The server cannot decrypt the blob. Wiped immediately upon delivery confirmation. Maximum retention: 30 days (text), 48 hours (media).
Salt	A 32-byte cryptographically random value generated by the GSK generator (Admin or Deputy) and broadcast to existing group members during GSK rotation. Combined with OLD_GSK via HKDF to produce NEW_GSK. Without the Salt, the old GSK cannot yield the new GSK.
Service Tunnel	A temporary, time-limited, configurable-capacity connection tunnel generated for professional service engagements. Defined by a time limit (default 30 days) and connection limit (default 1). Auto-collapses at expiry. Can be upgraded to a permanent bilateral tunnel.
SIM Binding	Hardware-level verification that links the physical SIM card (identified by IMSI) and mobile device (identified by IMEI) to a Shravu account instance. Mandated by India's DoT Telecommunication Cybersecurity Rules 2025, effective March 1, 2026.
Zero-Fill	The process of overwriting storage sectors with binary zeros to DOD 5220.22-M standard, making deleted data forensically irrecoverable. Used by Shravu's Poison Kill system and group member exit protocol.

4.2 Analysis Models

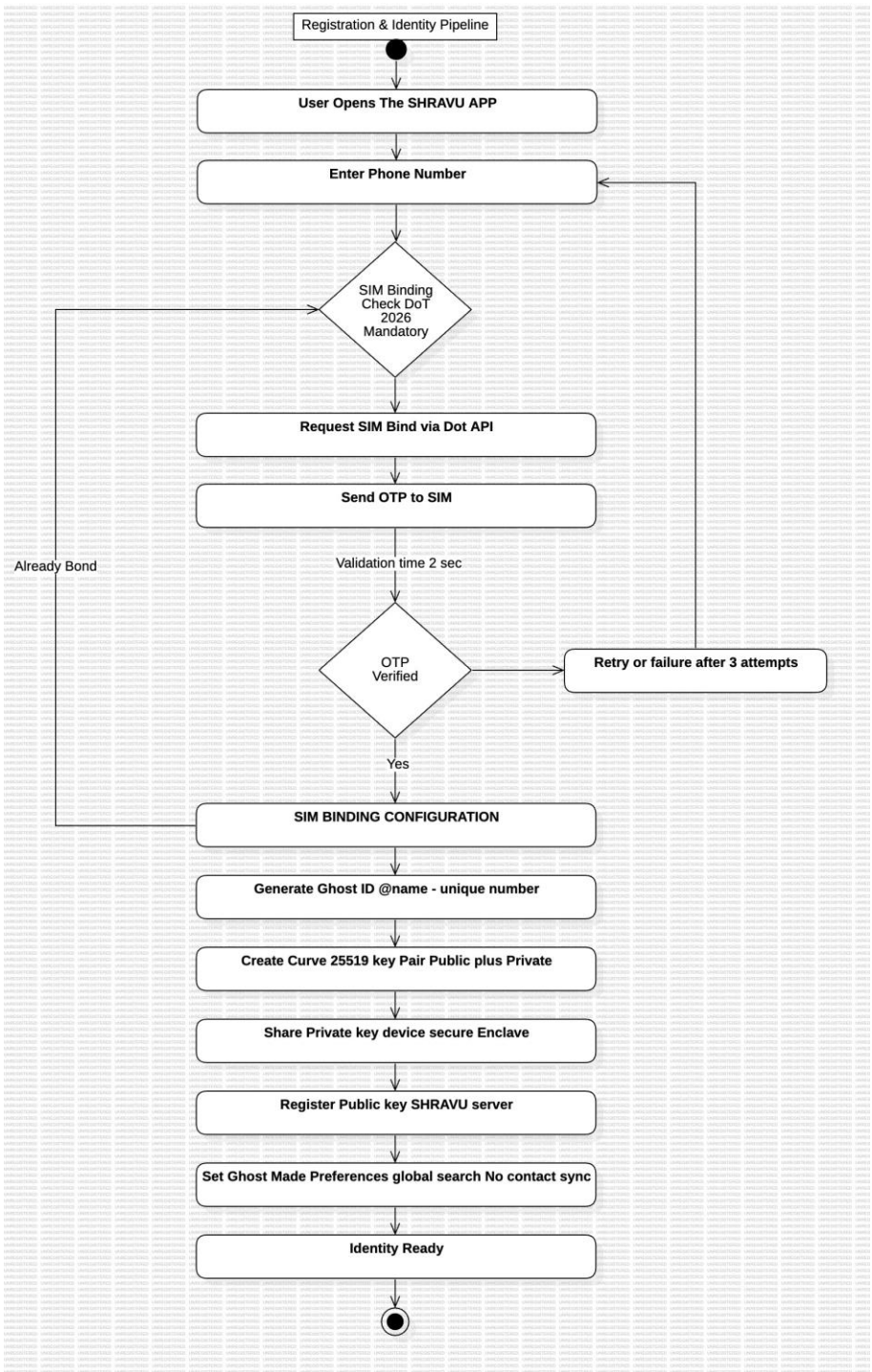
4.2.1 Use Case Diagram:



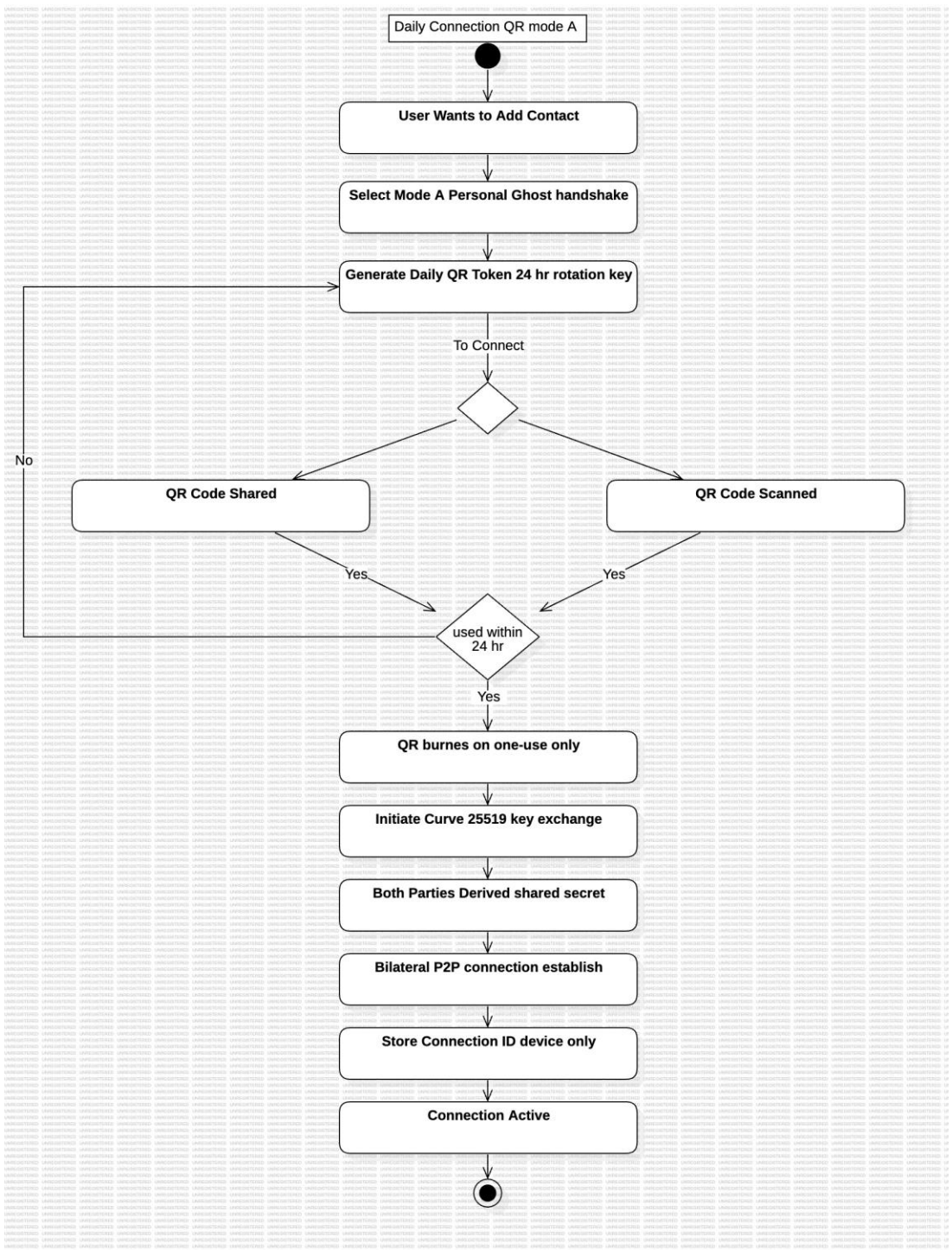


4.2.2 Activity Diagram:

- Registration & Identity Pipeline



• Daily Connection QR mode A



• **Daily Connection QR mode B**

Daily Connection QR mode B

Select Mode B Service ID QR

Configure parameter time limit or connection limit

Generate Service ID QR time-limited token

Contact scans QR?

Verify Within Limits

Timer or
Connection
limit
reached?

Reject Connection Limit Exceed

Service QR Expired Tunnel Closed

Establish Time-limited P2P Tunnel

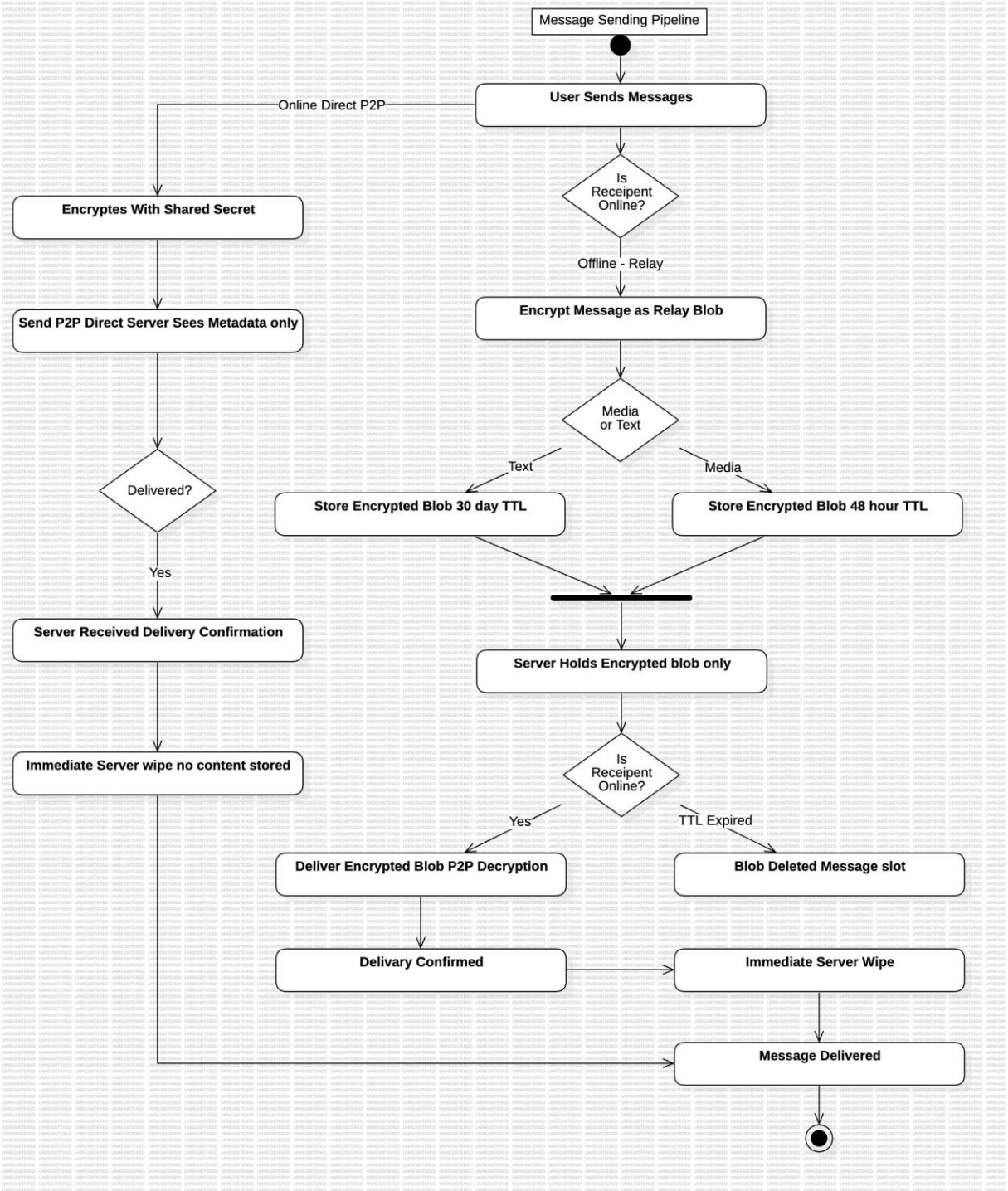
Profesional Tunnel With Expiry timer

Tunnel Expired?

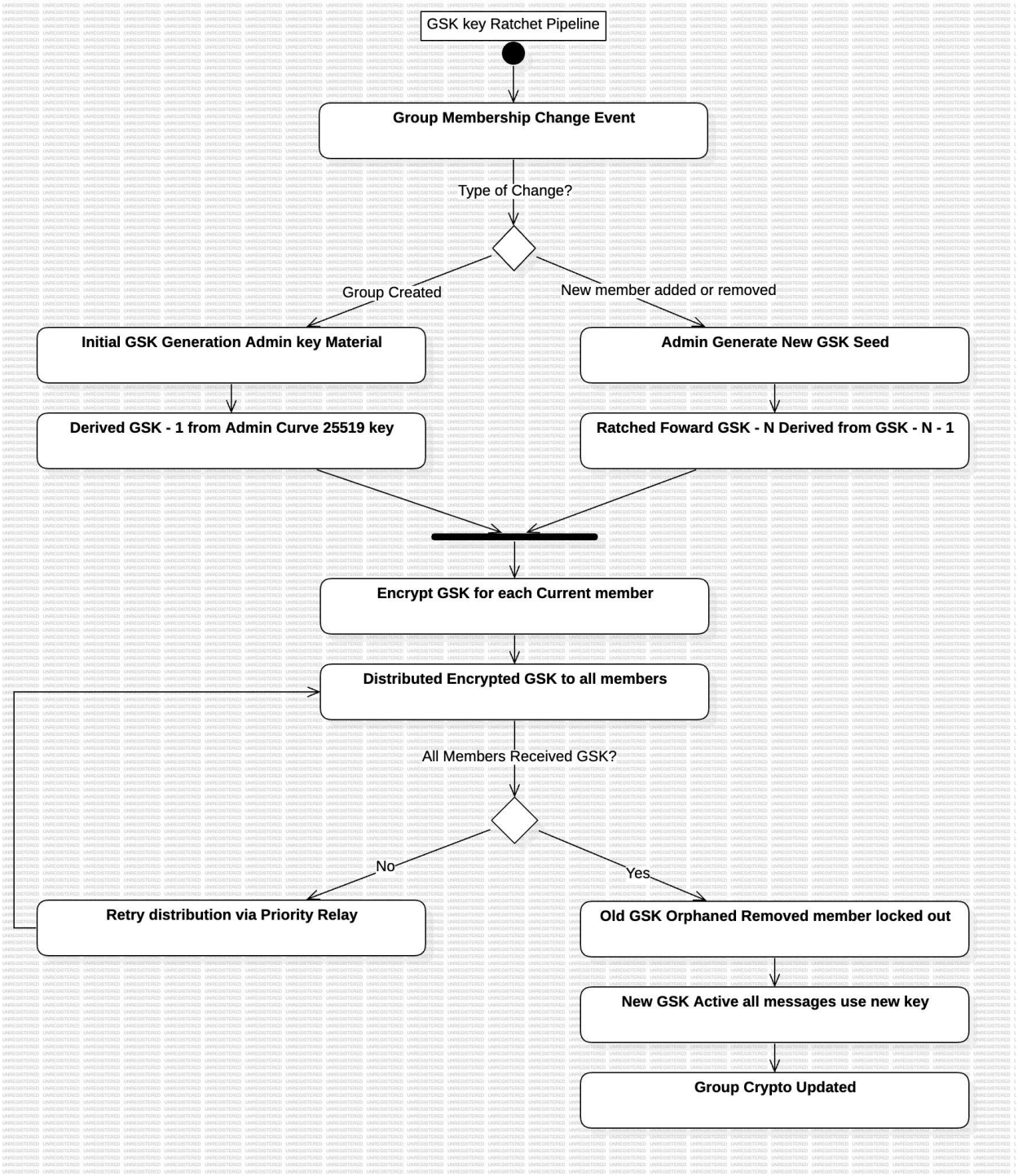
Connection Active

Tunnel Destroyed key wiped

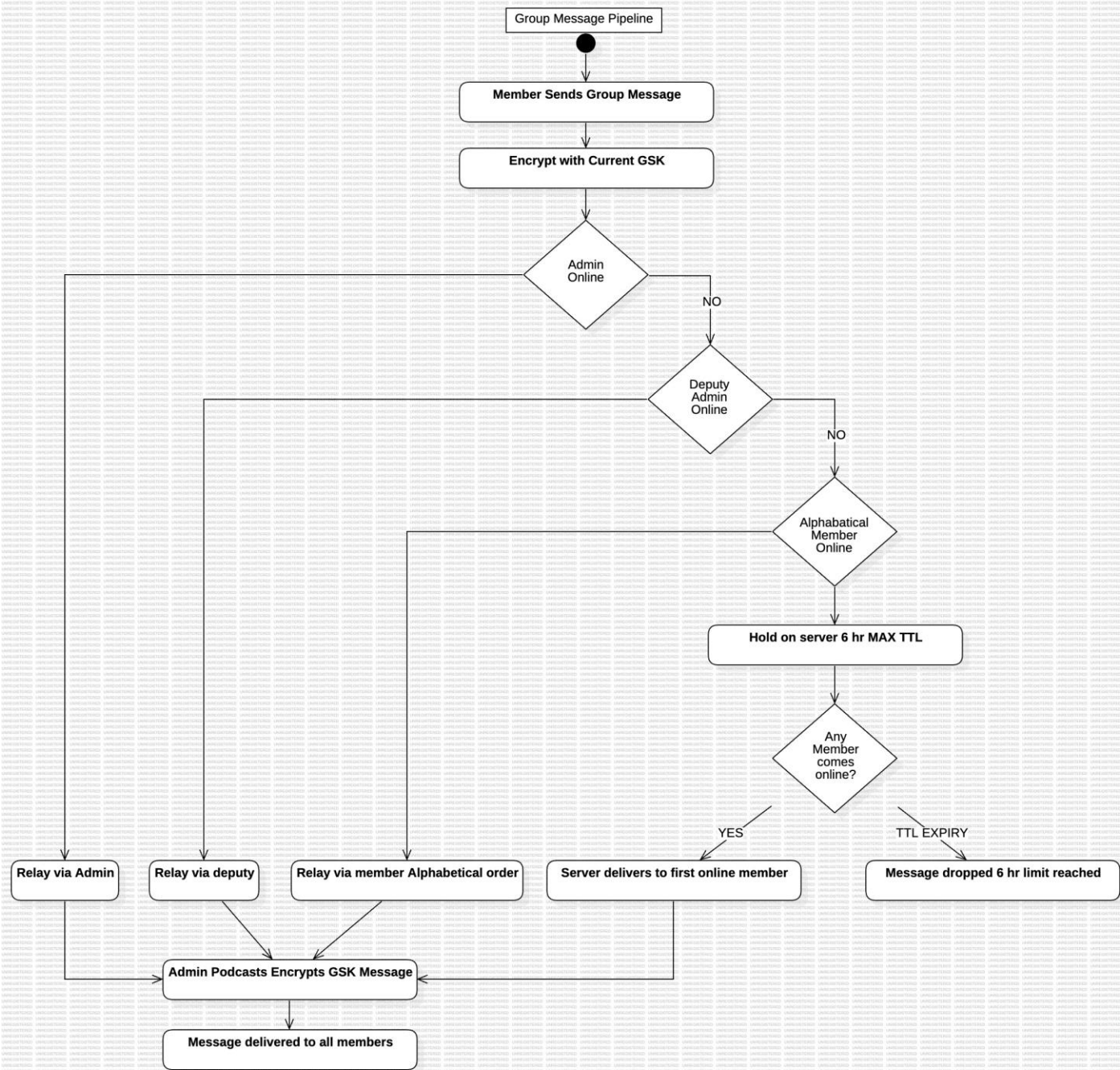
• Message Sending Pipeline



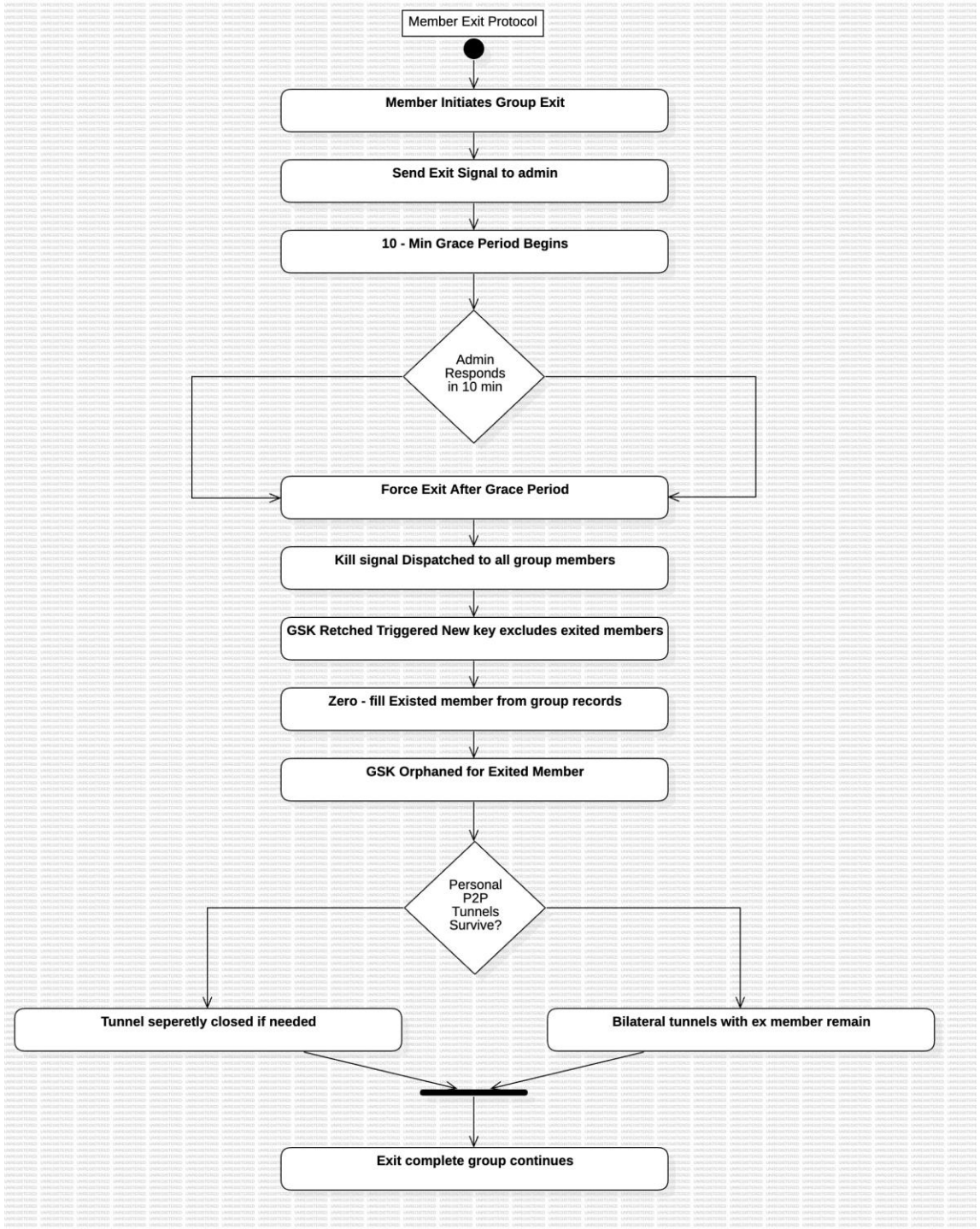
• GSK key Ratchet Pipeline



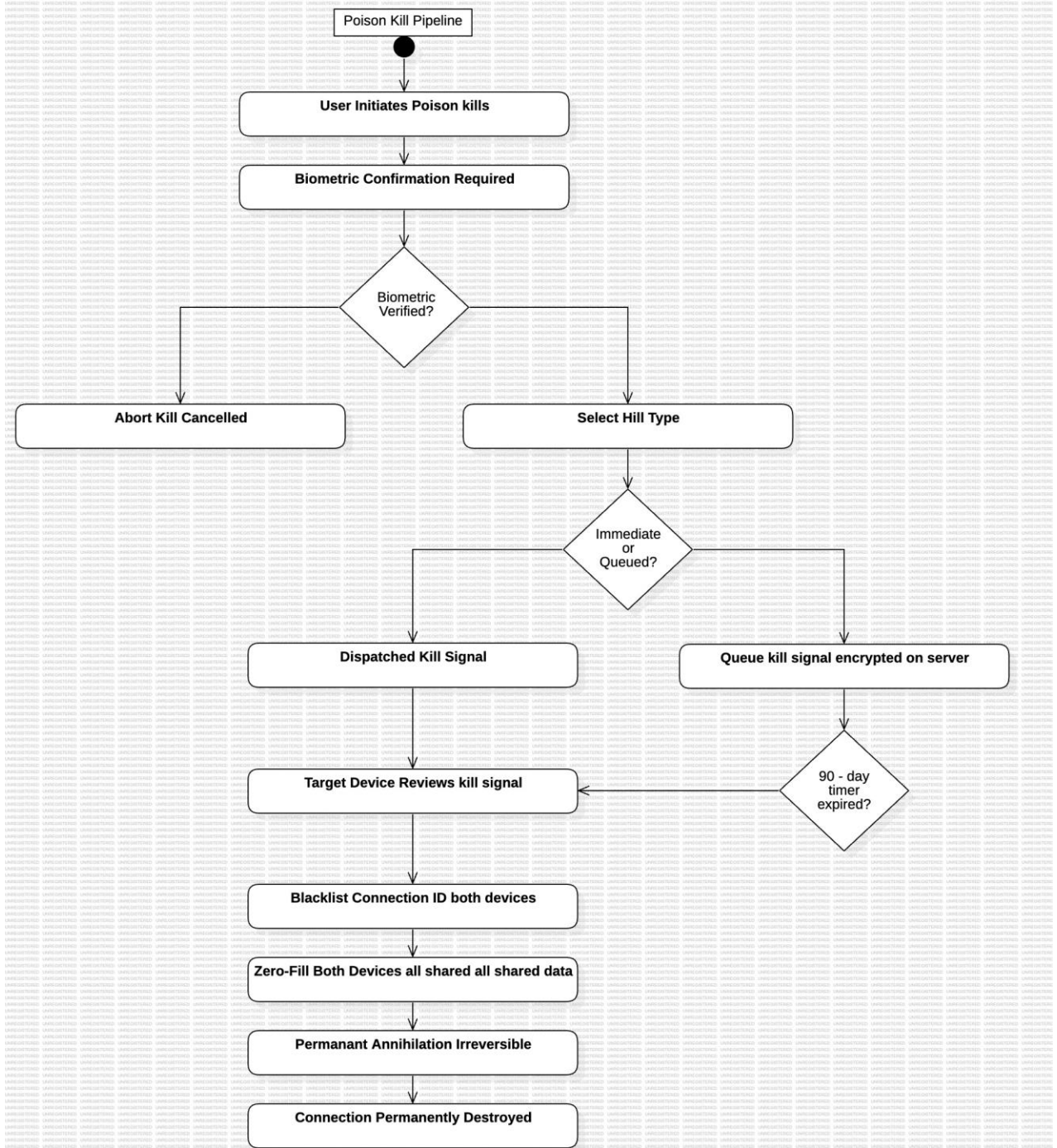
• Group Message Pipeline



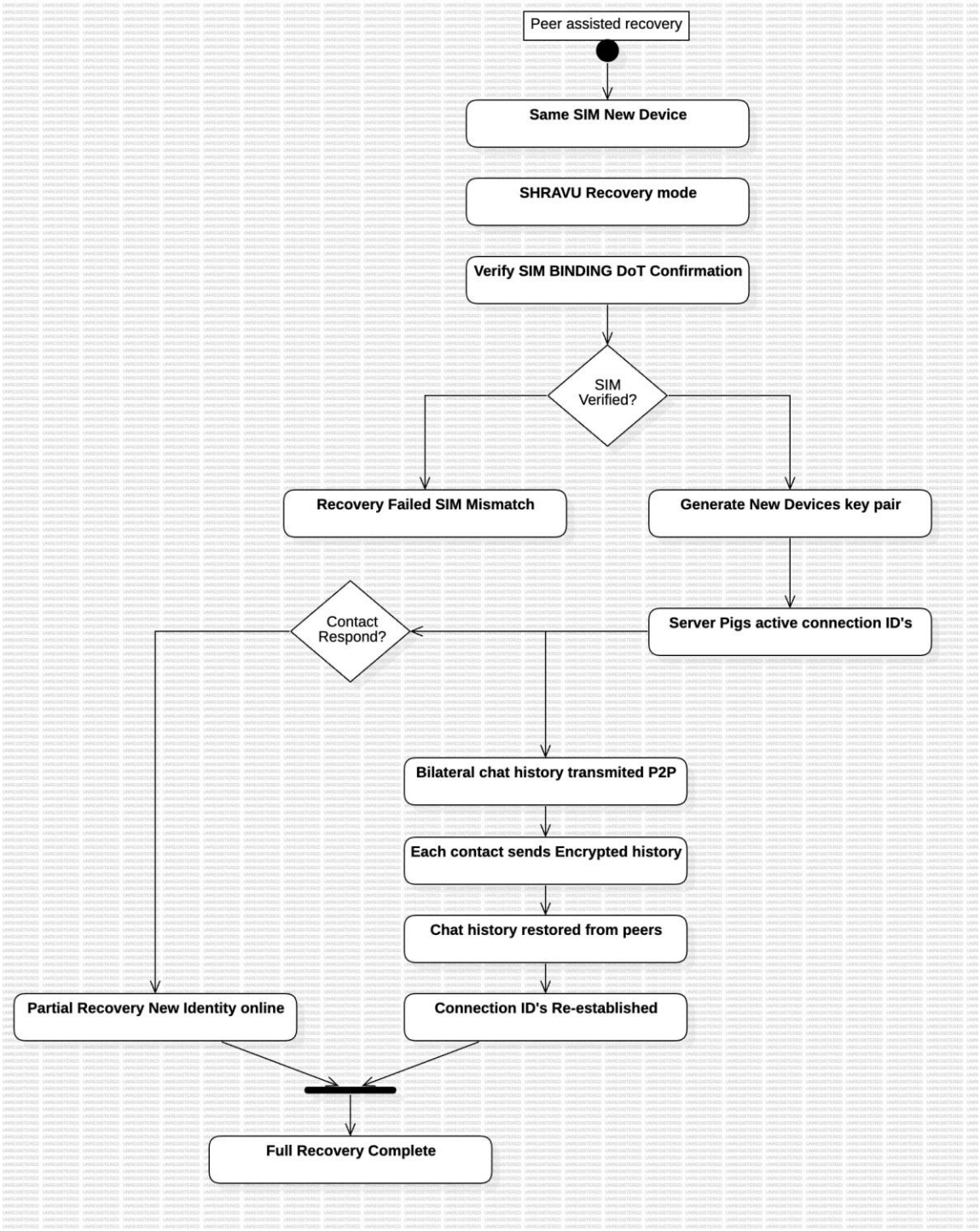
• Member Exit Protocol



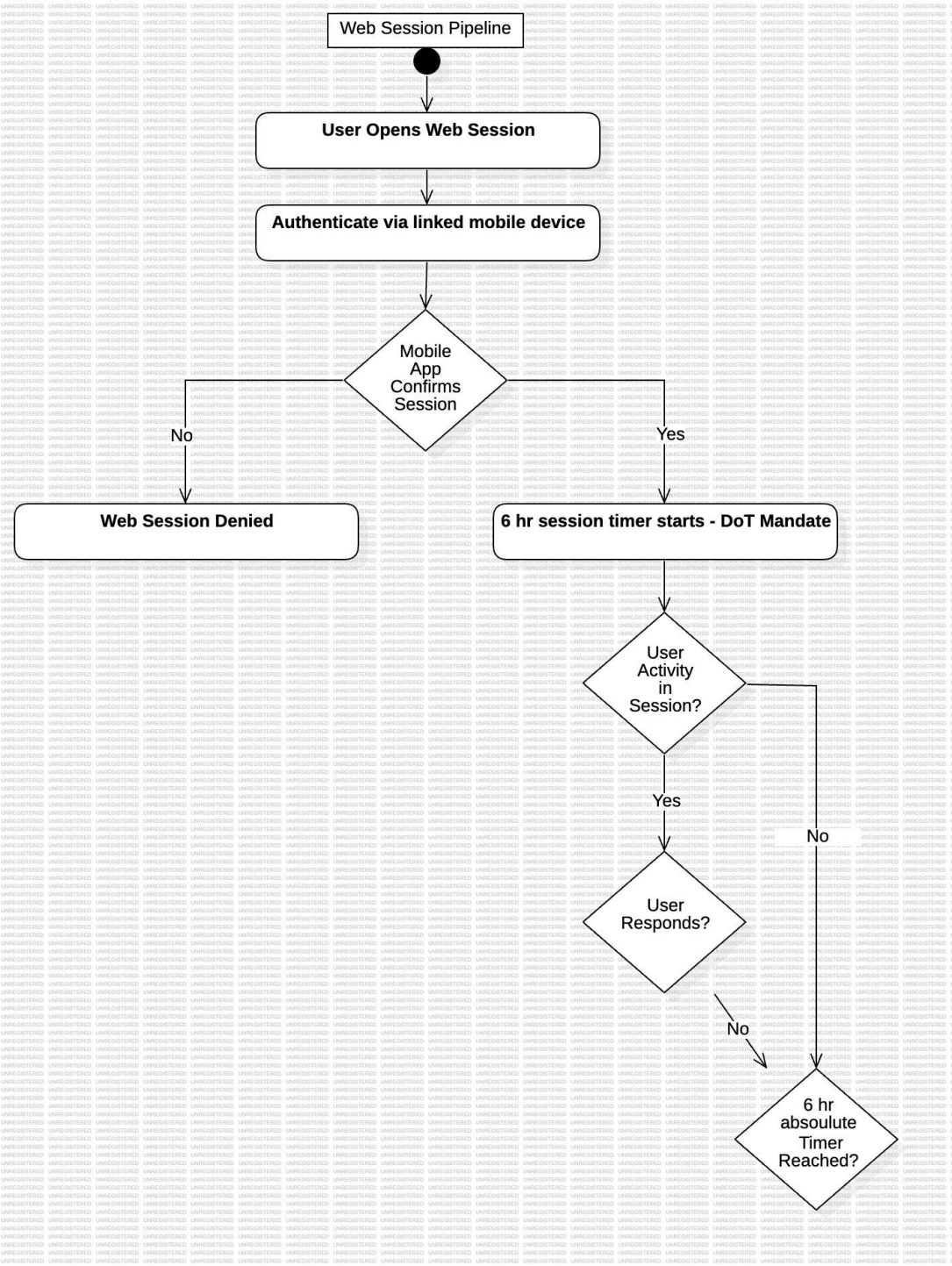
● **Poison Kill Pipeline**



• Peer assisted recovery



• Web Session Pipeline



4.2.3 Class Diagram:

